

1. Choose the PDE whose solution is $u(x, y)$ which satisfies

$$2u = (ax + y)^2 + b$$

where $p = \frac{\partial u}{\partial x}$ and $q = \frac{\partial u}{\partial y}$ and a, b are arbitrary constants.

- (a) $qx + py = p^2$
- (b) $px + qy = q^2$
- (c) $px + qy = p^2$
- (d) $qx + py = q^2$

Solution: [(b)]

Differentiating the expression that u satisfies with respect to x and also with y will give

$$2p = 2a(ax + y) \quad \text{and} \quad 2q = 2(ax + y)$$

which implies $a = \frac{p}{q}$. Substituting this expression in the later one will give

$$q = \frac{p}{q}x + y$$

which implies

$$px + qy = q^2$$

2. Choose the PDE whose general solution can be written as

$$x + y + z = f(x^2 + y^2 + z^2)$$

where f is the arbitrary function.

- (a) $(y - z)p + (z - x)q = x - y$
- (b) $(y - z)q + (z - x)p = x - y$
- (c) $(y - x)p + (z - y)q = x - z$
- (d) $(y - z)p + (z - x)q = x - z$

Solution: [a]

Differentiating partially with respect to x and y will give

$$1 + p = (2x + 2zp)f'(x^2 + y^2 + z^2)$$

$$1 + q = (2y + 2zq)f'(x^2 + y^2 + z^2)$$

Eliminating $f'(x^2 + y^2 + z^2)$ will give

$$\frac{1 + p}{2x + 2zp} = \frac{1 + q}{2y + 2zq}$$

which implies $(y - z)p + (z - x)q = x - y$

3. Let $u(x, y)$ be the solution of the given PDE then $u(1, 5) = \underline{\hspace{2cm}}$?

$$\left(\frac{\partial u}{\partial x}\right)^2 + \left(\frac{\partial u}{\partial y}\right)^2 = 0, \quad \forall (x, y) \in \mathbb{R} \times \mathbb{R}^+$$
$$u(x, 0) = 1, \quad \forall x \in \mathbb{R}$$

Solution: [1]

Now, the equation is true if and only if

$$\frac{\partial u}{\partial x} = 0 \quad \text{and} \quad \frac{\partial u}{\partial y} = 0$$

Solving the first equation will give $u(x, y) = f(y)$ for some function f . Substituting this in second equation will give

$$f'(y) = 0$$

which implies $f(y) = k$ for some constant k . Now, we have $u(x, y) = k$. But it is given that $u(x, 0) = 1$ for all x so it is only possible if $k = 1$. Hence $u(x, y) = 1$ for all $(x, y) \in \mathbb{R} \times (\mathbb{R}^+ \cup \{0\})$

4. Let $u(x, t) = U(x)T(t)$ satisfies the given 1d wave equation. Then which of the following is true?

$$\frac{\partial^2 u}{\partial t^2} = c^2 \frac{\partial^2 u}{\partial x^2}, \quad \forall x \in (0, L), t > 0$$
$$u(x, 0) = f(x), \quad \forall x \in [0, L]$$
$$u_t(x, 0) = g(x), \quad \forall x \in [0, L]$$
$$u(0, t) = 0, \quad \forall t \geq 0$$
$$u_x(L, t) = 0, \quad \forall t \geq 0$$

(a) $U(0) = 0$

(b) $U(L) = 0$

(c) $U'(0) = 0$

(d) $U'(L) = 0$

Solution: $[a, d]$

Substituting the expression of $u(x, t) = U(x)T(t)$ in the boundary conditions we get

$$U(0)T(t) = 0 \quad \text{and} \quad U'(0)T(t) = 0, \quad \forall t \geq 0$$

Since, we are looking for a non trivial solution u so $T(t) \neq 0$ for all t so $U(0) = U'(L) = 0$

5. If $f(x)$ and $g(x)$ are even functions in the following equation, then which of the following is true about the solution $u(x, t)$ of the equation?

$$\begin{aligned}\frac{\partial^2 u}{\partial t^2} &= c^2 \frac{\partial^2 u}{\partial x^2}, \quad x \in \mathbb{R}, t > 0 \\ u(x, 0) &= f(x), \quad x \in \mathbb{R} \\ u_t(x, 0) &= g(x), \quad x \in \mathbb{R}\end{aligned}$$

- (a) It is an even function in x
- (b) It is an odd function in x
- (c) It is not an even function in x
- (d) It is not an odd function in x

Solution: [(a)]

By D'Alembert's Solution

$$u(x, t) = \frac{f(x + ct) + f(x - ct)}{2} + \frac{1}{2c} \int_{x-ct}^{x+ct} g(s) ds$$

Now,

$$\begin{aligned}u(-x, t) &= \frac{f(-x + ct) + f(-x - ct)}{2} + \frac{1}{2c} \int_{-x-ct}^{-x+ct} g(s) ds \\ &= \frac{f(-(x - ct)) + f(-(x + ct))}{2} + \frac{1}{2} \int_{x+ct}^{x-ct} g(-z)(-dz) \\ &= \frac{f(x + ct) + f(x - ct)}{2} + \frac{1}{2c} \int_{x-ct}^{x+ct} g(z) dz \\ &= u(x, t)\end{aligned}$$

Hence, u is an even function in variable x .

6. Let $u(x, t) = X(x)T(t)$ be the non trivial solution of the given equation. Then which of the following is true about the k ?

$$\begin{aligned}\frac{\partial u}{\partial t} &= c^2 \frac{\partial^2 u}{\partial x^2}, \quad x \in (0, L), t > 0 \\ u(x, 0) &= f(x), \quad x \in (0, L) \\ u(0, t) &= 0 = u(L, t), \quad t > 0\end{aligned}$$

where $\frac{T'(t)}{c^2 T(t)} = \frac{X''(x)}{X(x)} = k$

- (a) Non trivial solution can not exist when k is non negative.
- (b) k can be a function of x or t
- (c) k can be a complex number

Solution: [(a)]

Since, in the equation of k left hand side is a function of t only and right hand side is a function of x only and both are equal so this is possible only if both expressions are equal to a constant function. As both functions are real valued so the coefficient has to be real. Therefore, k can not be a complex number.

Now, we know that the problem

$$X''(x) = kX(x)$$

with $X(0) = X(L) = 0$ has non trivial solution only when k is negative. So, from there one can conclude that only trivial solution exists when k is a non negative integer.